

Flutter UI Development Assignment

Objective

Implement a Flutter screen based on the provided design screenshot and animation reference video. The goal is to evaluate your ability to build production-quality Flutter applications following modern architectural patterns, clean coding practices, and performance optimization techniques.

Requirements

UI Implementation

Develop a Flutter screen that closely matches the provided design.

Expectations

- Pixel-perfect implementation.
 - Match layout, spacing, typography, colors, shadows, and visual hierarchy.
 - Proper handling of different screen sizes.
 - Support for portrait and landscape orientations.
 - Consistent UI across Android and iOS devices.
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Animation Implementation

Replicate the animations shown in the reference video.

Expectations

- Smooth transitions and interactions.
 - Appropriate animation curves and durations.
 - Use Flutter's animation framework efficiently.
 - Ensure animations work consistently across devices.
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Technical Requirements

Architecture

Implement the solution using **Clean Architecture** principles.

State Management

Use the **BLoC Pattern**.

Requirements

- Proper separation of Events, States, and Business Logic.
 - Avoid putting business logic inside Widgets.
 - State transitions should be predictable and testable.
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Dependency Injection

Use Dependency Injection.

Preferred options:

- GetIt
- Injectable
- Riverpod DI (if justified)

Requirements

- No direct dependency creation inside widgets.
 - All services and repositories should be injectable.
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Performance Optimization

Expected Practices

- Use const constructors where applicable.
- Avoid unnecessary rebuilds.
- Optimize widget tree structure.
- Use RepaintBoundary when necessary.
- Proper image handling and caching.
- Smooth 60fps animations.

Screenshots

Provide:

- Android screenshot

- iOS screenshot (if available)
 - Tablet screenshot (optional)
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Demo Video

Include a short demo video showing:

- UI implementation
 - Animation behavior
 - Responsiveness
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Submission

Submit:

1. GitHub Repository Link
2. APK/IPA (optional)
3. Demo Video
4. README Documentation